EE3-01
EE9-AC1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

MSc and EEE PART III/IV: MEng, BEng.and ACGI

Corrected copy

ANALOGUE INTEGRATED CIRCUITS AND SYSTEMS

Monday, 3 December 9:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : C. Toumazou
Second Marker(s) : P. Georgiou

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1. (a) Figures 1(a) and 1(b) show two popular biasing schemes typically used in analogue integrated systems. Explain the functionality of each of these circuits, highlighting the differences between the two.

[6]

(b) For the circuit of Figure 1(a), show that $\delta V_o / \delta T = 0$ (where T is temperature) if $(R_2/R_3)ln[I_1/I_2] = 29$ for $V_o = 1.283\ V$. Assume the temperature coefficient of V_{BE} to be -2.5mV/°C, the collector current of transistor Q_3 is 100μA and the device saturation current is $I_S = 1.2 \times 10^{-13} A$. Boltzmann's constant $k = 1.38 \times 10^{-23} J/K$ and the electron charge is $q = 1.6 \times 10^{-19} C$.

[8]

(c) For the circuit shown in Figure 1(b), show how the op-amp eliminates the voltage error between the Zener voltage and the output load to provide a regulated V_o.

[3]

(d) Sketch the circuit of a VGS multiplier and derive the equation of the output voltage.

[3]

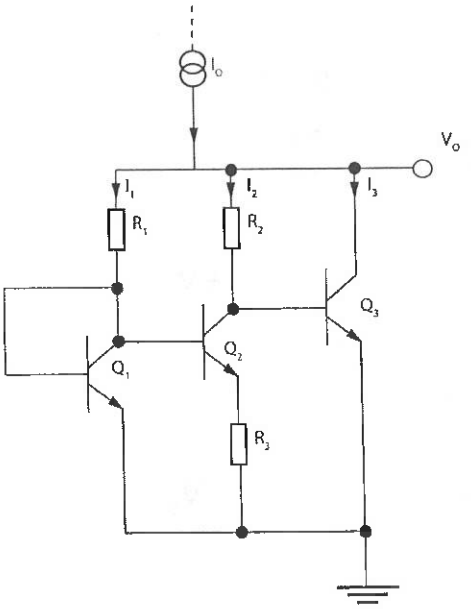


Figure 1(a)

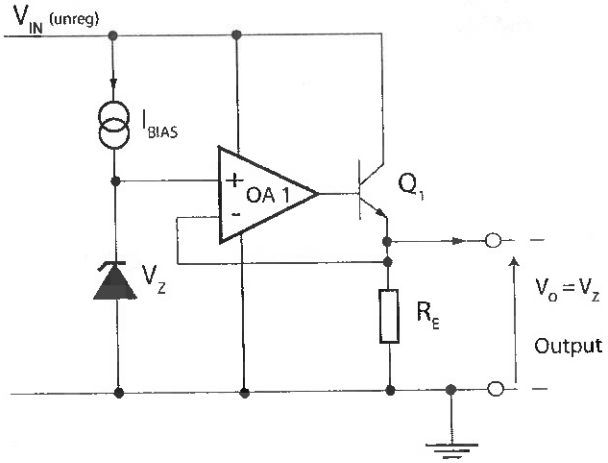


Figure 1(b)

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2. (a) Figure 2(a) shows a folded cascode connection. What is the main advantage of this design over the more classical cascode connection? Show via a brief sketch how the architecture of Figure 2(a) can be used to form the basis of a single stage fully differential folded cascode operational amplifier (op-amp) with common-mode feedback included.

[6]

(b) Figure 2(b) shows a two-stage CMOS op-amp. Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below. What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp?

[10]

(c) Explain why a resistor in series with the compensation capacitor C in Figure 2(b) can significantly improve the amplifier's phase margin.

[4]

CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	Kp ($\mu A/V^2$)	λ (V^{-1})	V_{T0} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

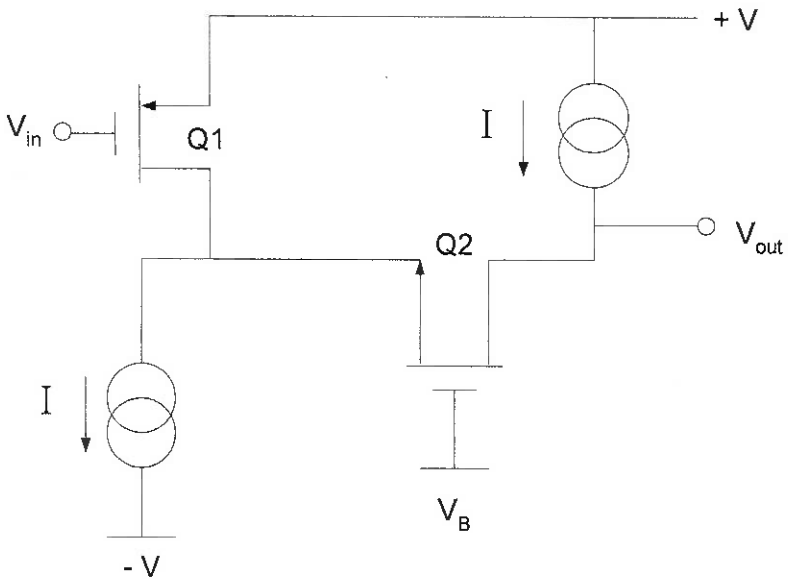


Figure 2(a)

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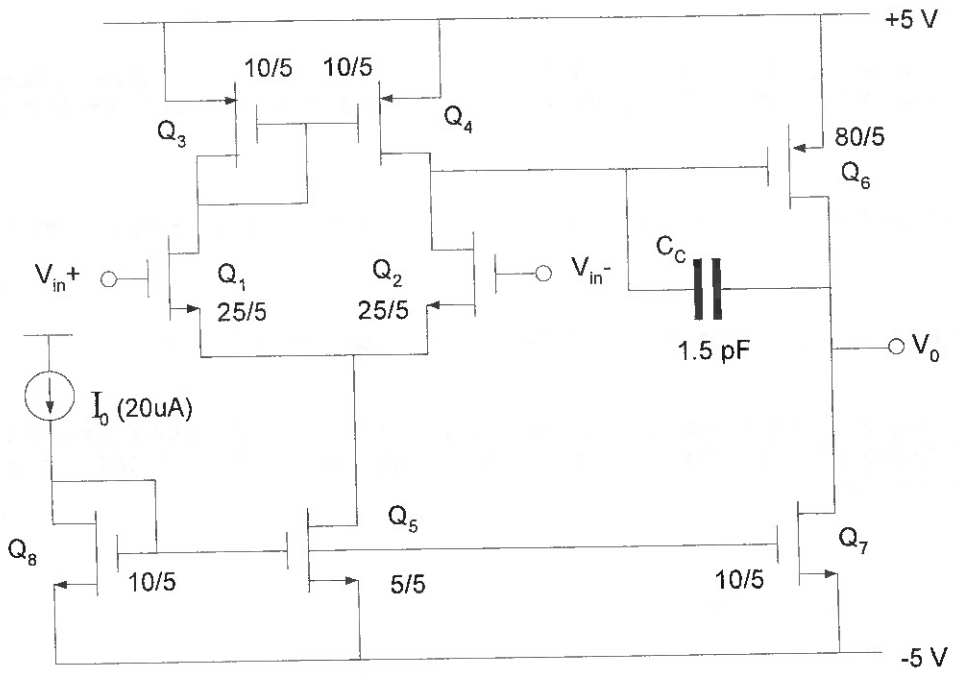


Figure 2(b)

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3. Figures 3(a) and 3(b) show the design of two differential precision integrators.

(a) Derive an expression for the time constant of each integrator. Assume that the sampled-data integrator of Figure 3(b) is driven by non-overlapping clocks and that the switches are ideal.

[6]

(b) Show how the integrator in Figure 3(b) can be made lossy, sketching the new schematic.

[3]

(c) Explain what noise source limits the dynamic range of sample data systems.

[3]

(d) The integrator in Figure 3(b) also includes an ADC. Describe the operation and sketch the architecture of a suitable ADC which overcomes this limitation in part 3.c with oversampling.

[8]

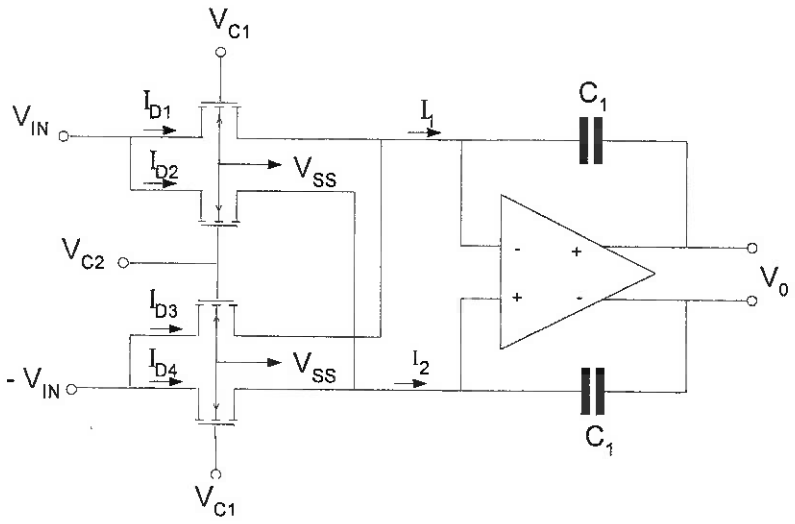


Figure 3(a)

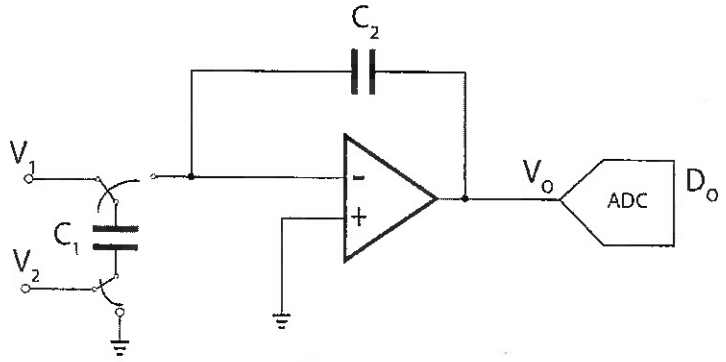


Figure 3(b)

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4.

(a) Sketch the basic design of a 3rd-order Chebyshev low pass switched-capacitor ladder filter.

[5]

The filter is to have a cut-off frequency of 10 kHz and assume a clocking frequency of 200 kHz. The values of the integration capacitor for the capacitor-based sections are 10.12 pF, and for the inductive based sections are 6.98 pF. All other switched capacitors are 1pF.

(b) From the circuit, estimate normalised passive component values for the original double-terminated LC prototype of the filter. All values should be normalised to 1 rad/s. You may assume that the clocking frequency is so high that the integrators can be assumed lossless.

[8]

(c) Explain the limitations of switch capacitor filters, showing a graph and a transfer function of how it's response deviates from an ideal integrator. Explain how this imposes a limitation on the clock frequency.

[5]

(d) Explain how the errors due to the limitations explained in part (c) be reduced.

[2]

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5. (a) Sketch the circuits of two supply independent current sources, including circuitry to guarantee their operation on start-up and explain their main differences.

[6]

(b) Explain the necessity of the automatic start up circuit for these topologies deriving the possible biasing points which exist.

[5]

(c) Using the equation for fractional temperature coefficient, explain which of the two circuits in 5.a is more insensitive to temperature.

[4]

(d) Describe how a PTAT could be used to bias long-tail pair such that it's transconductance becomes independent of temperature and derive the equation for the temperature independent transconductance.

[5]

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6. (a) Give one advantage and one disadvantage for each of the two CMOS current mirror circuits shown in Figures 6.1 and 6.2.

[4]

(b) For the current mirror of Figure 6.1, show how using a regulated cascode you can improve the output impedance stating equations for the output impedance before and after the improvement.

[6]

(c) For the current mirror of Figure 6.2 derive this voltage swing in terms of device threshold voltage V_T, clearly stating any assumptions you make.

[5]

(d) Discuss a suitable layout technique which would ensure the transistors in your current mirror are matched.

[5]

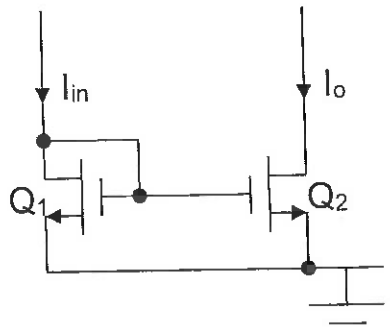


Figure 6.1

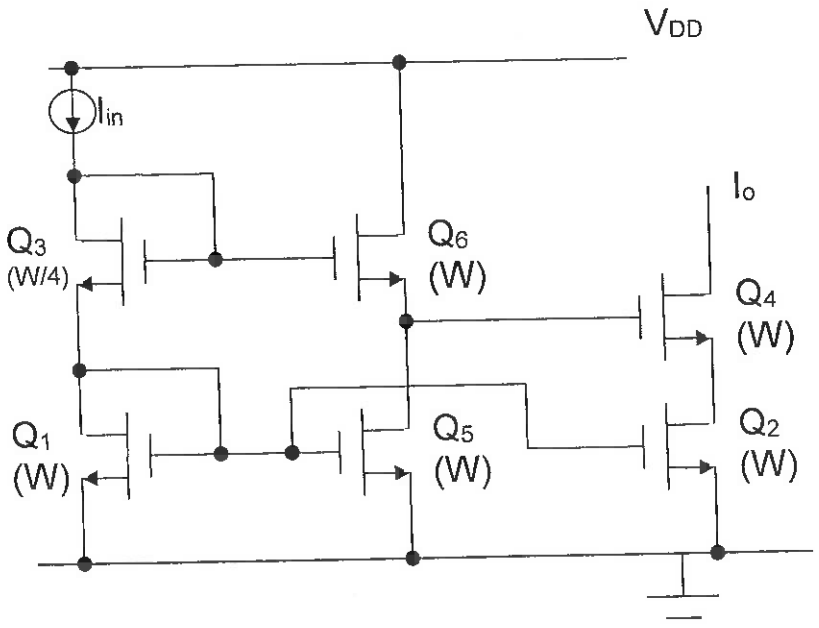


Figure 6.2

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